Functional Annotation Report: *Rtl8a* (Retrotransposon Gag-like 8A), Mouse

Target: *Rtl8a* / *Rtl8b* (Mus musculus) — UniProt **F6VWQ4 Protein:** Retrotransposon Gag-like 8 (RTL8; Mart/SIRH family of retrotransposon-derived genes; human orthologs are the FAM127/RTL8A–C genes) **Key domain:** RTL1/1-8/LDOC capsid-like domain (InterPro IPR032549; Pfam PF16297)

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1. Summary (Answer to the Research Question)

Rtl8a encodes **RTL8**, a small (~13 kDa) protein belonging to the **Mart/SIRH family of domesticated retrotransposon (Ty3/Gypsy-type) Gag-derived genes**. It is **not an enzyme, transporter, or classical signaling molecule**; rather, it is a **capsid-like protein-interaction adapter**. RTL8 corresponds to a truncated version of its paralog **PEG10**, consisting essentially of the **N-terminal Gag/capsid-like domain** and lacking the capacity to form particles on its own.

Its **primary molecular function** is to act as a **small structural adapter that regulates other retroelement-derived capsid proteins and the proteostasis machinery**, through two interlocking activities:

1. **Nucleolar targeting and control of UBQLN2 trafficking** — RTL8 localizes to **nucleoli** and promotes **nuclear translocation of the ubiquilin UBQLN2** into subnuclear protein-quality-control (PQC) compartments (validated in **mouse brain**).
2. **Antagonism of PEG10 capsid assembly** — RTL8 binds the **PEG10 capsid (Gag) domain**, is co-incorporated into PEG10 particles, and **inhibits PEG10 virus-like-particle (VLP) formation/release**, while also being the **required bridge** that lets UBQLN2 complex with PEG10 and be recruited to **stress granules**.

Thus RTL8 operates at the interface of **endogenous retrovirus-like protein biology and the ubiquitin–proteasome/stress-granule proteostasis network**, functioning in both the **nucleus/nucleolus** and the **cytoplasm (stress granules)**.

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2. Gene/Protein Identity Verification

The provided UniProt record (F6VWQ4; "Retrotransposon Gag like 8A/8B", mouse, *Rtl8a/Rtl8b*, RTL1/1-8/LDOC capsid-like domain) **matches** the published literature on **RTL8** (Mart/SIRH family; human orthologs = FAM127A/B/C, i.e. RTL8C/A/B):

- The literature describes RTL8 as a **Mart-family / retroelement-derived gene** with an **N-terminal Gag-like capsid domain** — exactly the IPR032549 / PF16297 domain annotated for F6VWQ4 [PMID 39775359].
- The UBQLN2–RTL8 interaction was confirmed **in mouse brain tissue**, directly matching the mouse organism of the target [PMID 35247097].
- RTL8 exists as a set of **near-identical paralogs** (in human RTL8A/B/C; in mouse *Rtl8a/Rtl8b/Rtl8c*), consistent with the dual gene naming (Rtl8a / Rtl8b) on this UniProt entry.

Conclusion: This is the correct gene. The literature describes the same protein family, domain, and organism. No evidence of a conflicting same-symbol gene was found.

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3. Molecular Identity and Structure

RTL8 is one of the smallest members of the **SIRH/Mart ("Sushi-ichi retrotransposon homolog" / Mammalian retrotransposon-derived) family** of domesticated **LTR-retrotransposon Gag genes**. Key structural facts:

- It is a **~13 kDa** protein [PMID 35247097].
- It **resembles a truncated form of PEG10**, sharing homology specifically with the **N-terminal Gag-like capsid domain** [PMID 39775359]. This is the IPR032549 "RTL1/1-8/LDOC capsid-like" domain annotated on F6VWQ4.
- **Alone, RTL8 cannot form virus-like particles (VLPs)** — unlike PEG10, which self-assembles into Gag-type capsids — but RTL8 **can be incorporated into PEG10-derived particles** [PMID 39775359].

Mechanistically this is important: because RTL8 is essentially a "lone capsid domain," it functions as a **capsid-interaction module** that docks onto, and co-assembles with, the full-length capsid proteins of its paralogs (notably PEG10), as well as onto proteostasis factors (UBQLN2). It has **no known catalytic activity**; its function is **structural/adaptor**, derived from the ancestral viral capsid fold.

Context: PEG10, the closest relative, is a Mart-family gene essential for placentation that forms Gag-type VLPs and even **binds and secretes its own mRNA** in extracellular vesicles [PMID 34413232]. RTL8 should be understood as a regulatory "satellite" of this capsid-forming machinery.

3.1 Evolutionary origin and domain architecture (bioinformatic evidence)

- The **UniProt record F6VWQ4** (gene *Rtl8a*, MGI:1913408) is a **113-amino-acid (~13 kDa)** protein consisting **almost entirely of a single "RTL1/3-8/LDOC capsid-like" domain (residues 18–113)**; the entry carries **no GO or subcellular-location annotations** (an uncharacterized TrEMBL entry), underscoring that functional knowledge comes from primary literature, not database annotation.
- RTL8 belongs to the family of **>50 cellular genes derived by recurrent "molecular domestication" of Gypsy/Ty3 LTR-retrotransposons** in mammalian genomes; these genes are **conserved across divergent mammals** and show similarity to **Gag**, the structural capsid protein of retroelements. Active Gypsy/Ty3 retrotransposons were themselves eliminated from the mammalian lineage **>180 million years ago** [PMID 19845640].
- Because the protein is a **lone Gag/capsid-like domain** (no reverse transcriptase, protease, or integrase), its biology is necessarily **structural/adaptor**, consistent with its experimentally demonstrated capsid co-assembly (PEG10) and protein-binding (UBQLN2) activities.

3.2 Mouse genomic context (Ensembl GRCm39)

The mouse genome contains a **tandem cluster of three near-identical *Rtl8* paralogs on the X chromosome**, all within <70 kb:

Gene	Ensembl ID	Location (chr X)	Strand
Rtl8c	ENSMUSG00000051851	52,609,752–52,612,251	+
Rtl8a	ENSMUSG00000067925	52,644,455–52,645,649	−
Rtl8b	ENSMUSG00000067924	52,671,141–52,680,104	−

Because *Rtl8a/8b/8c* encode **near-identical 113-aa proteins**, the UniProt entry F6VWQ4 is listed under both *Rtl8a* and *Rtl8b*, and most experimental assays detect "RTL8" collectively rather than resolving individual paralogs. The demonstrated RTL8 functions therefore apply to the cluster as a group, of which *Rtl8a* is a representative member. Note that the RTL8 cluster is **X-linked**, distinguishing it from its autosomal, imprinted relative *Peg10*.

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4. Primary Function and Mechanism

Because RTL8 is non-enzymatic, its "primary function" is best described as a **capsid-like adapter/regulator**. Two mechanistic axes are well supported:

4.1 Nucleolar protein and regulator of UBQLN2 (protein quality control)

- RTL8 was discovered as an interactor of **UBQLN2**, a ubiquitin-proteasome shuttle/adaptor protein implicated in ALS/FTD and other neurodegenerative diseases [PMID 35247097].
- Under both endogenous and overexpressed conditions, **RTL8 localizes to nucleoli** [PMID 35247097].
- When co-expressed, **RTL8 promotes nuclear translocation of UBQLN2**, including during **heat-shock stress**, delivering UBQLN2 into **ubiquitin-enriched subnuclear structures containing PQC components** [PMID 35247097].
- This activity is **specific to UBQLN2** (not the related UBQLN1 or UBQLN4), and **UBQLN2 reciprocally stabilizes RTL8** protein levels in human cell lines and mouse brain [PMID 35247097].

Interpretation: RTL8 is a nucleolar adapter that governs the **nuclear/subnuclear distribution of UBQLN2**, thereby feeding into the **ubiquitin–proteasome protein-quality-control pathway**. This links the gene to clearance of aberrant nuclear proteins and to neurodegeneration-relevant proteostasis.

4.2 Antagonist of PEG10 capsid (VLP) assembly

- RTL8 **binds the N-terminal (capsid) domain of PEG10** and is **packaged into PEG10 particles** [PMID 39775359].
- RTL8 co-expression **decreases secreted PEG10 VLP abundance** while **increasing intracellular PEG10**, indicating it **inhibits PEG10 VLP formation or release** [PMID 39775359].

- Modulating RTL8 levels changes PEG10-derived VLP abundance in cells that naturally produce them [PMID 39775359].

Interpretation: By co-assembling into PEG10 capsids through the shared Gag fold, RTL8 acts as a **dominant-negative–like structural inhibitor (antagonist)** of PEG10 particle assembly/secretion — a rare example of one domesticated retroelement protein regulating another.

4.3 Required bridge for the UBQLN2–PEG10–stress-granule network

- **RTL8 confers on UBQLN2 the ability to complex with and regulate PEG10** [PMID 40680123].
- PEG10, a **core stress-granule component** that forms VLPs within granules, **recruits UBQLN2 to stress granules under stress — but only when RTL8 is present** [PMID 40680123].
- Changing RTL8/PEG10/UBQLN2 levels **remodels stress-granule disassembly kinetics and translation recovery** [PMID 40680123; preprint

P 39484508

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Interpretation: RTL8 is an **obligatory adapter** coupling UBQLN2 proteostasis machinery to PEG10 capsids within **cytoplasmic stress granules**, integrating two endogenous retrovirus-like proteins into the cellular stress response.

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5. Subcellular Localization

RTL8 operates in **multiple compartments**, consistent with its adapter role:

Compartment	Evidence
Nucleolus (steady state)	RTL8 localizes to nucleoli endogenously and when overexpressed [PMID 35247097]
Nucleus / subnuclear PQC bodies	Co-translocates with UBQLN2 into ubiquitin-enriched subnuclear structures [PMID 35247097]
Cytoplasm / stress granules	Required for UBQLN2 recruitment to PEG10-containing stress granules [PMID 40680123]
PEG10 particles / VLPs	Incorporated into PEG10-derived capsid particles [PMID 39775359]

6. Pathways and Biological Context

- **Ubiquitin–proteasome system / protein quality control:** via control of UBQLN2 localization and stability [PMID 35247097].
- **Stress-granule dynamics and translational recovery:** via the RTL8–PEG10–UBQLN2 axis [PMID 40680123].
- **Endogenous retrovirus-like (Mart/SIRH) protein network:** direct regulation of PEG10 capsid assembly [PMID 39775359].
- **Tissue context:** RTL8 is **broadly expressed in many of the same tissues as PEG10, including brain** [PMID 39775359]; the functional interaction with UBQLN2 was confirmed in **mouse brain** [PMID 35247097], making the brain a particularly relevant site of action.
- **Disease relevance:** UBQLN2 and PEG10 are implicated in **ALS/FTD and other neurodegenerative diseases**; RTL8, as their shared regulator, is positioned within this neurodegeneration-associated proteostasis network [PMID 35247097;

P 40680123

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7. Supported and Refuted Hypotheses

Supported: - H1: RTL8 is a domesticated retrotransposon Gag/capsid-like protein (truncated PEG10-like). ✔ [PMID 39775359] - H2: RTL8 is a nucleolar protein that controls UBQLN2 nuclear trafficking. ✔ [PMID 35247097] - H3: RTL8 regulates PEG10 capsid/VLP assembly (antagonist). ✔ [PMID 39775359] - H4: RTL8 bridges UBQLN2–PEG10 interaction in stress granules. ✔ [PMID 40680123]

Refuted / not supported: - The protein is **not** an enzyme or transporter; no catalytic or transport activity is described. Its "retrotransposon Gag-like" name reflects evolutionary origin, not an active transposition function.

8. Limitations and Future Directions

- Much of the mechanistic detail comes from **human cell lines and overexpression**; the strongest mouse-specific evidence is the **UBQLN2–RTL8 interaction in mouse brain** [PMID 35247097]. Endogenous mouse *Rtl8a*-specific loss-of-function phenotypes are not yet well characterized in the retrieved literature.
- Mouse has **near-identical paralogs (*Rtl8a/8b/8c*)** arranged as a **tandem X-linked cluster** (chr X ~52.61–52.68 Mb, Ensembl GRCm39), complicating isoform-specific functional assignment; most antibodies/assays detect "RTL8" collectively.
- The field is young (key papers 2022–2025); **structural confirmation** (e.g., cryo-EM of RTL8 within PEG10 capsids) and **in vivo developmental/neurological roles** in mouse remain open.
- Future work: conditional *Rtl8a* knockout phenotyping, isoform-resolved expression mapping, and direct tests of whether RTL8-regulated PEG10/UBQLN2 proteostasis modifies neurodegeneration models.

References (PMIDs)

- PMID **19845640** — Volff, J.N. Cellular genes derived from Gypsy/Ty3 retrotransposons in mammalian genomes (2009; evolutionary origin review).
- PMID **35247097** — RTL8 promotes nuclear localization of UBQLN2 to subnuclear PQC compartments (2022; mouse brain validation).
- PMID **39775359** — The gag-like gene RTL8 antagonizes PEG10-mediated virus-like particles (2024).
- PMID **40680123** — Endogenous retrovirus-like proteins recruit UBQLN2 to stress granules and shape their functional biology (2025).
- PMID **39484508** — (preprint of the above) Endogenous retrovirus-like proteins recruit UBQLN2 to stress granules (2024).
- PMID **34413232** — Mammalian retrovirus-like protein PEG10 packages its own mRNA / VLP context (2021).